CENG381 Stochastic Processes

Rate of Convergence - Question Set 1

Q1 (30 points). Consider the 2-state Markov chain on {E, W} with transition matrix:

```
         E      W
    E [  0.7    0.3 ]
    W [  0.4    0.6 ]
```

a) Find the stationary distribution pi = (pi_E, pi_W).

b) Define Delta_t = mu_t(E) - pi_E. Show that Delta_{t+1} = (1-p-q) * Delta_t, where p = 0.3 and q = 0.4, and identify the rate of convergence lambda_2 = 1-p-q.

c) If the chain starts at E (mu_0(E)=1), how many steps are needed until |Delta_t| < 0.01? (Use the formula Delta_t = lambda_2^t * Delta_0.)

Q2 (35 points). A Markov chain on {0, 1} has transition matrix P with eigenvalues lambda_1 = 1 and lambda_2 = 1 - p - q. It can be diagonalized as P = S^(-1) D S, where D = diag(1, lambda_2).

a) For p = 0.2, q = 0.5, find D and write P in diagonalized form.

b) Use P = S^(-1) D S to derive an explicit formula for P^n, the n-step transition matrix.

c) Compute P^(33) for the entry p_{EE}^(33). Simplify using the formula p_{EE}^(n) = q/(p+q) + lambda_2^n * p/(p+q).

Q3 (35 points). A lazy frog sits on two lily pads. To make the chain converge faster, we introduce a 'lazy' version: at each step, with probability 1/2 the frog stays in place (does nothing), and with probability 1/2 it follows the original transitions from matrix P below.

```
         E      W
    E [  0.6    0.4 ]
    W [  0.3    0.7 ]
```

a) Write the transition matrix Q of the lazy chain as Q = (1/2)I + (1/2)P.

b) Find the eigenvalues of Q. How does lambda_2(Q) compare to lambda_2(P)?

c) The rate of convergence of the lazy chain is |lambda_2(Q)|. Is the lazy chain faster or slower to converge than the original? Why might one still prefer the lazy chain in practice (hint: think about periodicity)?